nature energy

Review article

https://doi.org/10.1038/s41560-026-01969-z

Challenges, technological pathways and trade-offs of perovskite solar modules for long-term operation

Received: 4 May 2025

Accepted: 2 January 2026

Published online: 09 March 2026

Check for updates

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Perovskite solar modules (PSMs) have emerged as a promising photovoltaic technology due to their high efficiency, low fabrication cost and compatibility with lightweight and flexible applications. However, ensuring long-term reliable performance under real-world conditions remains a critical barrier to commercialization. PSMs degrade through mechanisms that differ substantially from those affecting established technologies such as silicon, particularly under environmental stressors like ultraviolet light, oxygen, temperature cycling and reverse bias. Here we provide an analysis of the degradation pathways specific to perovskite modules and discuss why standard accelerated tests often fail to predict outdoor performance. We conceptualize challenges across material, device and module levels and evaluate strategies to mitigate ion migration, interfacial breakdown and encapsulation failure. By highlighting the need for realistic testing protocols and durable materials, we propose a framework highlighting key challenges, technological pathways and the trade-offs required to extend perovskite module lifetimes towards long-term operation, aiming to guide the development of PSMs capable of a 30-year operational lifetime.

Perovskite solar modules (PSMs) have emerged as a promising technology in the field of photovoltaics (PVs) due to their excellent photoelectric properties and low-temperature solution processing—over 27% for single-junction cells and over 34% for perovskite–silicon tandem solar cells. The state-of-the-art module efficiency has exceeded 21% for pre-pilot-size modules, according to the National Laboratory of the Rockies (NLR) champion module efficiency chart[1]. However, long-term operational lifetime remains a critical barrier to commercialization. In striking contrast to the tremendous amount of indoor stability test results, reports on the module-level outdoor performance of PSMs under diverse climatic conditions, never mind long-term (>2 years) and system-level datasets, remain scarce. So far, no public demonstration has reported perovskite modules with aperture efficiencies above 18% maintaining T_{90} (the time a device maintains 90% of its initial performance under specific test conditions) beyond three years of continuous outdoor operation. Understanding the degradation mechanisms specific to PSMs, such as ion migration, electrochemical reactions and mechanical fatigue under prolonged thermal, electrical, optical and environmental stress, is therefore crucial to achieving durable, field-stable devices.

One of the primary differences between PSMs and traditional solar technologies, such as Si or cadmium telluride (CdTe) PVs, lies in the intrinsic material properties of the perovskite absorber layer[1]. Unlike Si, which is in the form of covalently bonded robust crystals, perovskites are composed of soft ionic lattices with lower formation energies (0.1–0.5 eV per atom, compared to 2–4 eV per atom for Si), making them

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more susceptible to environmental stressors such as light, heat and mechanical stress[2,3]. The water-soluble nature of perovskites makes them very sensitive to moisture[4]. Despite these challenges, the defect tolerance of the perovskite electronic structure allows these materials to maintain high performance even in the presence of imperfections that would be detrimental in Si or CdTe[5]. Moreover, the soft lattice and ionic nature of halide perovskites open unique possibilities for post-fabrication healing, lattice reorganization and interface engineering, offering prospects for long-term resilience not readily available in traditional semiconductors. For example, the 'self-healing' effect has been observed when degraded perovskite devices were exposed to light cycling[6]. At the same time, the ionic features underpin major degradation pathways. Interactions between perovskites and adjacent functional layers, mainly driven by van der Waals forces, add notable challenges to module encapsulation and degradation associated with thermal cycling. Various electrochemical reactions involving cations (such as formamidinium, FA^+) and anions (such as iodide, I^-) are accelerated by external stressors such as reverse bias and ultraviolet (UV) illumination, leading to perovskite decomposition[7]. Moreover, the inherently facile ion migration gives rise to halide phase segregation, lattice distortion and eventual amorphization under prolonged operational stress[8]. In PSMs, degradation often initiates from interconnections and interfaces between layers, particularly those formed during laser scribing and other fabrication steps where defects, imperfections and compromised encapsulation can accelerate the degradation.

To better frame the discussion, degradation pathways can be broadly classified into intrinsic and extrinsic factors. Intrinsic degradation arises directly from material properties and device architecture (for example, ionic migration, lattice instability or defect chemistry). Extrinsic degradation involves external influences, such as moisture ingress, oxygen penetration, mechanical stress and encapsulation failures. Intrinsic factors set fundamental limits to durability, but extrinsic factors often accelerate or trigger these intrinsic pathways. Understanding their interplay is crucial: improvements in intrinsic degradation pathways can reduce the sensitivity to extrinsic stresses, while robust encapsulation and packaging can substantially mitigate intrinsic vulnerabilities. Explicitly separating these categories and recognizing their interactions helps guide targeted strategies for improving perovskite module reliability.

Predicting the operational lifetime of PSMs from accelerated stress tests is challenging due to their limited reproducibility and complex degradation kinetics. The Arrhenius model is commonly used to describe temperature-dependent degradation, assuming a constant activation energy (E_a) and rate constant (k): $k(T) = A e^{-E_a/RT}$, where A is the pre-exponential factor, R the gas constant and T the absolute temperature. However, PSMs rarely follow simple degradation kinetics. Their degradation typically proceeds through multiple non-Arrhenius regimes, as summarized in Box 1.

In this Review we first compare indoor tests and outdoor operation, emphasizing the need for realistic testing protocols. We then discuss the impact of critical stressors, including blue light and oxygen exposure, and the combined effects of light and heat on device durability. We also examine specific strategies to mitigate reverse bias effects at both the device and module levels. Finally, we explore critical challenges in scaling perovskite cells to module dimensions, focusing on uniformity, interconnections and robust encapsulation techniques essential for long-term reliability.

Limits of indoor tests for predicting outdoor performance

Initial studies of PSCs under constant illumination and inert environments showed promising durability, evolving from hundreds to thousands of operational hours[9]. Despite encouraging results, these controlled tests do not fully replicate real-world conditions[10]. Limited outdoor studies, particularly for modules, highlight discrepancies between indoor performance and outdoor reliability. A handful of studies using indoor testing suggest operational lifetimes of decades; however, these lifetimes have not been realized in real-world testing. Outdoor conditions introduce complexities such as varying temperatures, light intensity and spectrum, humidity and oxygen conditions, as well as mechanical stresses, which are challenging to simulate effectively in indoor environment tests. As shown in Fig. 1a, a key discrepancy arises from the spectral mismatch between indoor and outdoor illumination sources, where some simulated light often lacks the UV component present in sunlight. Additionally, Fig. 1b shows that the stability of indoor light sources (for example, xenon, metal halide, light-emitting diodes (LEDs)) can vary over time, further complicating reliable lifetime testing. The combined impact of these factors results in substantially different degradation behaviours, as illustrated in Fig. 1c, where outdoor-tested perovskite devices degrade much faster than those tested indoors. The consequences of these discrepancies are evident at the module level. Although test-transferability is a general PVs challenge, perovskite solar cells (PSCs) are especially affected due to their multiphase degradation modes and metastability, as discussed in Box 1. Work is needed to validate qualification standards and stress testing protocols for perovskite modules and panels[9].

UV durability

Although humidity can be mitigated by appropriate packaging[11], other critical factors influencing the field performance of PSCs, such as UV exposure, thermal cycling and light cycling, require thorough investigation. Indoor stress tests commonly use LED lights without a UV spectrum, whereas sunlight contains ~5% UV. Fei and colleagues emphasized the necessity of including UV in indoor testing, noting accelerated outdoor degradation that was largely due to UV exposure[12]. If devices lack a UV-blocking layer, indoor tests should incorporate the UV spectrum to realistically represent outdoor conditions. Although some devices may exhibit minimal sensitivity to UV, validation through testing remains essential. Jiang and colleagues demonstrated minimal UV influence in specific systems, suggesting that indoor LED tests could serve as a viable option, but confirmation is critical[13]. To mitigate UV-driven degradation, the most effective strategies combine UV-filtering encapsulants, photo-stable hole-transport layers, and compositional tuning towards mixed-halide or Cs-enriched perovskites. These improve resistance to photon-induced halide segregation while maintaining optical transparency[14,15].

Light and temperature cycling

Compared to UV exposure, the impact of light and temperature cycling is substantially more complex. Indoor tests typically use constant illumination at constant temperature to expedite performance evaluations, omitting the day–night cycle occurring outdoors[7]. However, perovskite devices often experience beneficial 'light soaking' effects, which temporarily boost efficiency by enabling self-healing or the wake-up process, as described in Box 1. Outdoor tests commonly show differences in device performance between morning and evening (Box 1), highlighting the importance of light and temperature cycling[16,17]. Studies by Khenkin and colleagues indicate that the use of constant illumination and temperature inadequately simulates outdoor conditions, potentially inducing nonrepresentative degradation pathways[18,19]. However, generalizing these effects remains challenging due to the extensive variability in perovskite compositions, device architectures and device fabrication processes. Figure 1d illustrates typical outdoor light cycling profiles, characterized by strong variations in irradiance and temperature during day–night cycles that are absent from standard indoor testing. To fully understand and accurately simulate real-world stress conditions, further research is essential, particularly in module configurations, as many initial studies are on individual small-area cells. Although combined-stress testing enables faster prediction of outdoor stability, it may also trigger nonrepresentative failure modes,

BOX 1

Non-Arrhenius degradation in PSCs

Wake-up. Power conversion efficiencies (PCEs) increase when devices are aged under light, heat or a combination thereof, or without any stress and simply in dark storage. Aging processes involve interface reconstruction, crystallization and improved passivation. It should be noted that wake-up is not unique to perovskites. It happens across other PV technologies as well, including Si[115], CdTe and copper indium gallium selenide (CIGS)[116], which shows PCE increases of ~1–5%. However, in PSCs, a huge PCE increase of >200% is sometimes observed after a few minutes of light soaking[117]. Wake-up is often described in terms of open circuit voltage (V_{OC}) and fill factor (FF), or FF only. Measuring under continuous illumination, for example, at light soaking condition (ISOS-L), does not capture wake-up behaviour. There are multiple extensive studies on the origin of performance wake-up under light soaking and temperature[118], pointing to possible origins such as continued grain growth[119], a redistribution of ions and ionic defects, as well as light-induced lattice expansion-related crystallinity improvement[120], though the detailed mechanisms remain unclear.

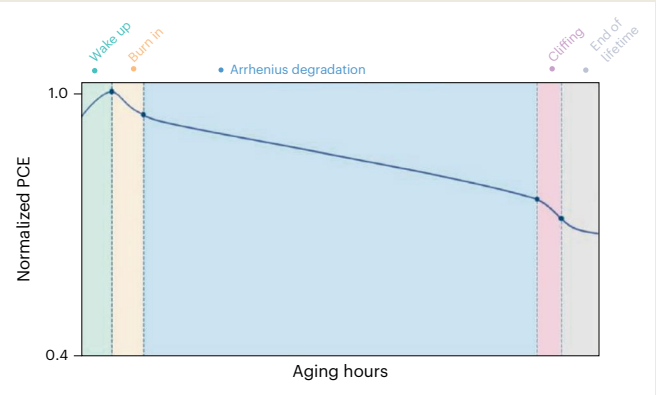
Burn-in. Performance wake-up may be followed by rapid performance degradation, commonly known as burn-in, when devices are stressed under both outdoor and indoor accelerated testing[121]. Although the burn-in process sometimes mirrors the wake-up behaviour in opposite, where V_{OC} and FF primarily dictate the decay dynamics of PCE, in many cases, the short circuit current density (J_{SC}) becomes the dominant factor in PCE degradation dynamics. Such different dynamics in PCE decay indicate different mechanisms for the burn-in process. The wake-up and burn-in processes could take place simultaneously, with one dominating based on the physical or chemical process during aging, though this is not well understood.

Arrhenius degradation. This is the longest stage in solar cell degradation and features mechanisms that differ from those during wake-up and burn-in. It is a quasi-linear regime where the degradation rate, k, appears constant over hundreds to thousands of hours and is temperature-dependent, enabling empirical E_a extraction, although overlapping mechanisms often cause deviations. Several recent indoor studies showed a single E_a across the operating temperature range, as well as a constant k at a given temperature[122,123], but this does not always hold true. There could be different mechanisms of perovskite degradation that have not yet been understood[124]. The Arrhenius degradation follows the Arrhenius equation, assuming that the degradation rate is not limited by drift or diffusion, and that Arrhenius-based modelling is valid only over time intervals where the degradation proceeds linearly and the concentration of reacting species remains constant (see the figure)[125,126].

Cliffing. This is an abrupt collapse in performance at long aging times that is linked to mechanical or interfacial failure (for example, delamination, corrosion or cracking)[127]. If a device has a very short lifetime, it is hard to distinguish between burn-in and cliffing.

End of lifetime. Ultimately, irreversible chemical, mechanical and interfacial failures dominate, leading to a severely reduced PCE that is practically inconsequential.

These overlapping regimes demonstrate that no single activation energy or rate constant can describe PSC lifetime. Although the Arrhenius relation remains useful within the linear degradation phase, real devices exhibit coupled, time-dependent processes that demand broader non-Arrhenius frameworks to accurately predict long-term stability.



Box Fig. 1 | An overview of indoor light + heat accelerated performance dynamics. Arrhenius-type degradation applies only during time intervals where the degradation proceeds linearly, and the concentration of reacting species remains constant.

illustrating the trade-off between accelerated data acquisition and physical relevance[20].

Degradation pathways induced by blue light and oxygen

Although UV and light/temperature cycling are often absent in indoor testing, blue light and oxygen are present under some test conditions, particularly in devices with encapsulation. However, current encapsulation strategies remain insufficient for full mitigation of these two stresses. Blue light, an important component of the visible spectrum, cannot be entirely blocked or down-converted without efficiency loss or increased costs. It has been shown to accelerate ion migration, phase segregation and defect formation, while also introducing additional degradation pathways[21]. Another unsolved instability issue comes from oxygen-related degradation. Previous studies, such as those reported in refs. 22–24, have demonstrated that edge sealants such as desiccated polyisobutylene (PIB) can effectively prevent moisture ingress for over 20 years; however, they are less effective at blocking oxygen, which can penetrate the seal within weeks to months, depending on the temperature (Fig. 2a)[25]. Similarly, encapsulants utilizing a polyolefin (POE) base polymer have limited oxygen barrier properties[26].

Blue light durability

Blue light was reported by Wang and colleagues to induce more severe degradation in perovskite materials compared to the effect of lower-energy photons (Fig. 2b)[27]. High-energy blue photons can dissociate organic ammonium cations, as reported by Nickel and colleagues[28]. Additionally, blue light can be absorbed by PbI_2 (with a bandgap energy of $E_g \approx 2.4$ eV), facilitating the photooxidation of iodide (I^-) into molecular iodine (I_2) via trapped holes. Once generated, I_2 can quickly damage perovskites. Even in the absence of illumination, I_2 readily binds with I^- to form triiodide (I_3^-), which reacts strongly with organic cations, leading to deprotonation of the organic cations and the regeneration

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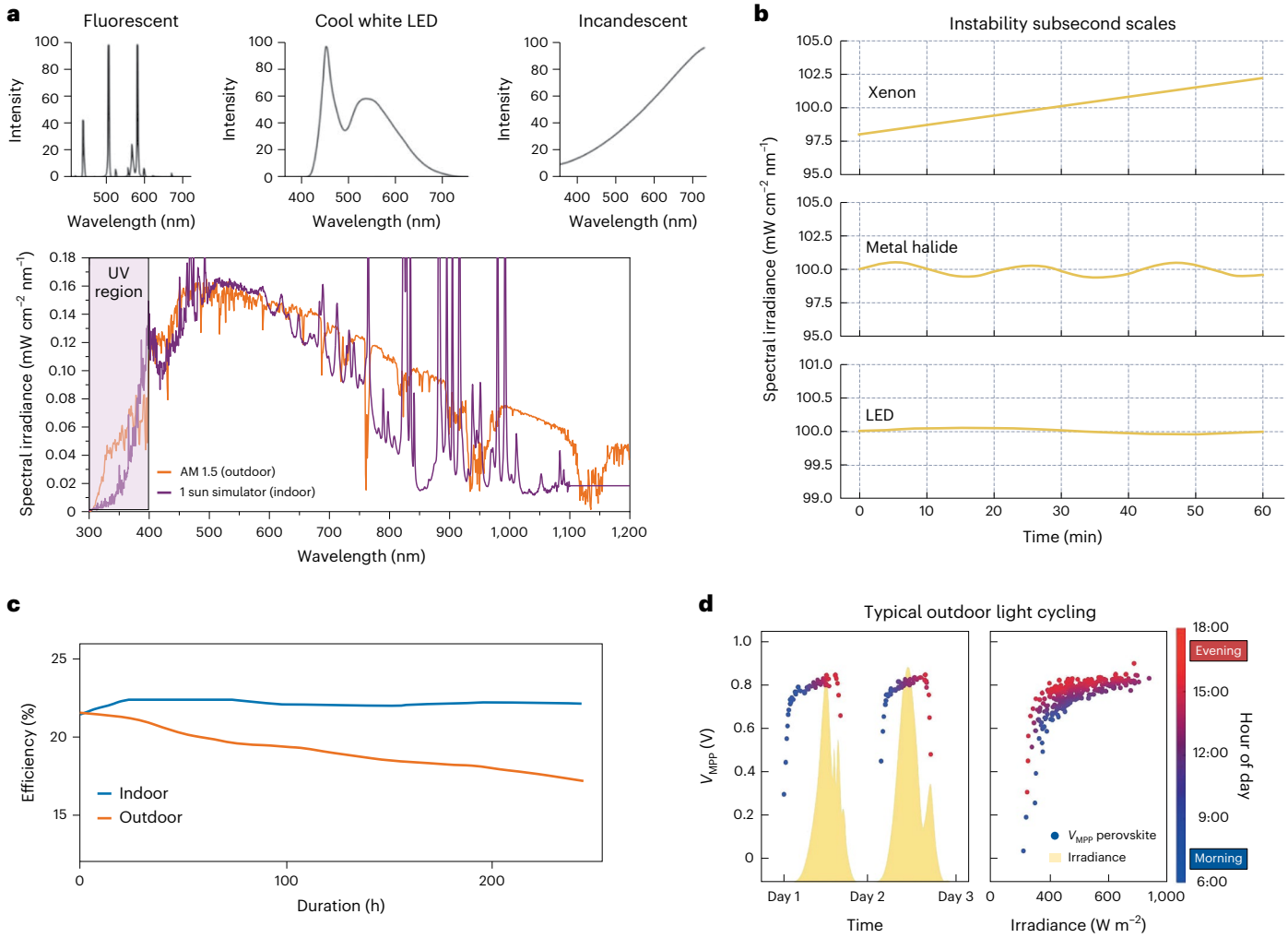

Fig. 1 | Discrepancies between indoor testing and outdoor operation for PSCs and modules. a, Comparison of the spectral irradiance of typical indoor light sources (fluorescent, LED and incandescent) and the AM 1.5 outdoor solar spectrum with a typical indoor solar simulator spectrum. **b**, Short-term spectral instability of different indoor solar simulators (xenon, metal halide and LED) over a 60-min duration. **c**, Efficiency degradation comparison of PSCs under controlled indoor testing and outdoor operational conditions, highlighting the accelerated degradation outdoors[12]. The indoor testing of devices was carried out under an LED lamp (100 mW cm^{-2} with <0.1% UV inside, 60° ± 5 °C, open circuit (OC) conditions), and those tested outdoors were under sunlight and ~20 °C average daytime temperature (North Carolina, USA). The light intensity for testing indoors was 100 mW cm^{-2}, and the data were collected from between six and eight devices for each group. **d**, Representative example of typical outdoor light cycling, demonstrating variations in irradiance, temperature and humidity over multiple day–night cycles, conditions that are challenging to replicate accurately indoors. V_{MPP}, maximum power point voltage. Adapted from ref. 114, CC BY 4.0.

of I_2/I_3^- (ref. 29). This process initiates a degradation cycle wherein I_2/I_3^- perpetuates structural breakdown. Furthermore, the absorption of blue light by I_3^- accelerates the deprotonation of organic cations by generating highly reactive iodine radicals ($I_2^{\cdot-}$ and $I^{\cdot}$), as discussed by Ren and others[30]. The photolysis of I_2 also generates I^- radicals, which react with I^- to form $I_2^{\cdot-}$ (ref. 31).

Several strategies have been explored to mitigate blue-light-induced degradation by reducing or eliminating PbI_2 and I_2/I_3^- formation within perovskite films. Common approaches, such as that of Chen and colleagues, include using PbI_2-deficient nonstoichiometric perovskite compositions, avoiding oxidized precursors containing I_2/I_3^-, and incorporating reductive additives to suppress I_2/I_3^- formation during ink preparation, device fabrication or operation[32]. Previous studies, including those by Ren and colleagues, have introduced radical scavengers such as perfluorodecyl iodide, which can capture iodine radicals before they contribute to further degradation[33]. However, these reductive additives or radical scavengers generally suffer from two limitations: their effectiveness depends on their spatial distribution within the films, and they may deplete over time without a self-regenerating mechanism. Strategies such as defect passivation and improved crystallization processes have also been reported to reduce defect density, thereby delaying the photogeneration of PbI_2 and I_2/I_3^-. Interestingly, reports from Yang and colleagues have suggested that intentionally incorporating I_2/I_3^- into perovskite precursors can reduce defect density and improve film material stability[34,35]. Further investigations are necessary to determine the trade-offs between the potential benefits and detrimental effects of I_3^- incorporation.

Oxygen durability

The role of oxygen in perovskite degradation has also been widely studied[36]. Under dark conditions or short-term exposure, oxygen alone causes minimal damage, if not passivating, in perovskites, which allows for ambient fabrication[37]. In the simultaneous presence of 1-sun illumination and an ambient environment, photooxidation accelerates perovskite degradation rates by approximately one order of magnitude compared to dark, inert conditions, leading to PbI_2 formation within tens of hours[38–40]. The extent to which oxygen-induced degradation contributes to efficiency loss during long-term operation, relative to other

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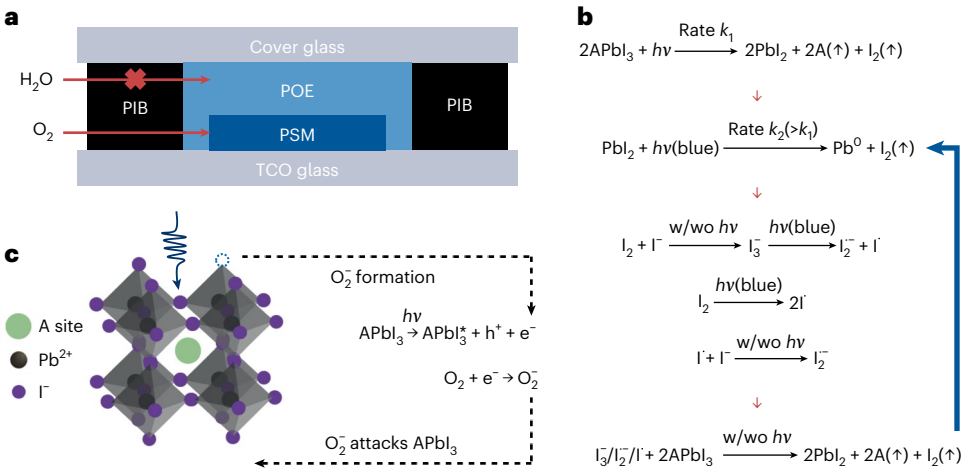

Fig. 2 | Illustration of the degradation modes of perovskites exposed to oxygen and blue light. a, Schematic showing that the current PIB/POE encapsulation method cannot block oxygen ingress. **b**, Mechanism of blue-light-induced perovskite degradation. **c**, Permeation of O_2 in the perovskite (arrow). The schematic shows oxygen reacting with the perovskite to form reactive superoxide species, preferentially at defect sites, damaging the perovskite.

degradation pathways, remains an open question. A well-documented mechanism involves oxygen interacting with photoexcited perovskites to form highly reactive superoxide species (O_2^-), which subsequently drive perovskite decomposition (Fig. 2c)[41–43]. Defects within the perovskite lattice have been identified as key reaction centres for O_2^- formation, introducing mid-gap states that act as carrier traps[39,44–46]. Thus, strategies aimed at reducing defect density, such as surface passivation with iodide salts, have been reported by Aristidou and colleagues to enhance resistance to oxygen-related degradation[39]. Additional approaches, including the rapid extraction of photogenerated electrons before they interact with oxygen or the introduction of superoxide scavengers, have also been proposed to suppress O_2^- formation[47]. However, they face the same limitation as iodine scavenging.

The charge-transport materials (CTMs) used in PSMs, particularly organic semiconductors, are also known to be vulnerable to oxidation. Their degradation pathways have been extensively characterized in the context of organic PVs[48]. However, further studies are required to determine the relative contribution of perovskite oxidation compared to the degradation of other functional layers in the long-term performance of PSMs. Future work could involve selectively passivating the perovskite layer while independently stabilizing the charge-transport layers, followed by comparative aging studies under operational conditions to isolate and quantify their respective degradation pathways.

Emerging encapsulation approaches, such as atomic layer deposition (ALD)-grown metal oxides, offer promising solutions. Several approaches have been developed to reduce oxygen penetration into the perovskite layer. These include multilayer encapsulation structures combining polymer barriers (for example, PIB-based edge seals) with inorganic gas diffusion barriers such as ALD-deposited aluminium oxide (Al_2O_3) or sputtered silicon oxide (SiO_x)[49,50]. Recent work has shown that hybrid structures with alternating inorganic/organic layers can achieve water vapour transmission rates (WVTRs) of $<10^{-6}$ g m^{-2} d^{-1} while maintaining optical transparency and mechanical flexibility[51]. For example, flexible perovskite modules encapsulated using ALD–polymer hybrid stacks demonstrated strong resistance to oxygen-induced degradation under damp heat and light soaking[52,53]. Furthermore, additives or interlayers at the perovskite–contact interface (for example, metal-chelating molecules or oxygen-scavenging polymers) can reduce the impact of residual oxygen that diffuses through the barrier system[54]. However, a techno-economic analysis is necessary to determine whether prioritizing oxygen barrier improvements is feasible for enhancing long-term device durability[55].

Effects of combined light and heat stress

It is generally accepted that testing under illumination and elevated temperature (for example, 85 °C) is needed to screen for early PSC failures, but this test alone may not detect abrupt degradation events (for example, cliffing)[13]. The best reported thermal-light performance involves multi-cation PSCs demonstrating T_{90} lifetimes of over 3,000 h at 90 °C under LED illumination and maximum power point tracking conditions[27]. However, achieving a similar performance using perovskite minimodules with efficiencies over 18% remains challenging, with no report so far on this size. Here we review and analyse the degradation pathways of PSCs and modules under combined thermal (−40 °C to 85 °C) and 1-sun illumination conditions, focusing on the chemical, interfacial and physical processes that govern long-term stability. This will help us uncover the remaining research gaps.

Physical degradation of perovskites

Metal-halide perovskites have thermal expansion coefficients roughly ten times higher than substrates such as transparent conductive oxides (TCOs) and glass[56]. At elevated temperatures, perovskite films experience substantial internal strain, altering bond lengths and angles and potentially increasing the number of defects[57,58]. However, strain alone is usually not the main degradation factor; combined heat and illumination greatly accelerate ion migration, phase transformations and phase segregation—all unique to perovskites compared to incumbent PV technologies. High bromide content (>25%) strongly induces phase segregation, although the impact of lower bromide percentages remains debated[59]. Performance data typically favour pure iodide compositions, such as $FA_xCs_{1-x}PbI_3$, although some mixed halide compositions also demonstrate acceptable high-temperature durability[60].

A-site cation segregation in mixed-cation perovskites was initially argued to be extremely slow or nonexistent due to the much larger activation energy barrier. However, there is evidence that this segregation happens when the solar cells are stressed for longer times and at a higher temperature under light[32]. Figure 3a illustrates the phase transition process, showing a pathway from a metastable black phase to a non-perovskite yellow phase, although the phase transformation in $FA_xCs_{1-x}PbI_3$ perovskites is still not well understood. A-site cation migration is often accompanied by perovskite decomposition, and understanding the detailed process of the mixed-cation degradation can help in finding methods to stabilize it. Mixed FA/Cs compositions containing 2–15% Cs exhibit the stable black α-phase up to 85 °C, whereas pure FA perovskites undergo partial δ-phase conversion above

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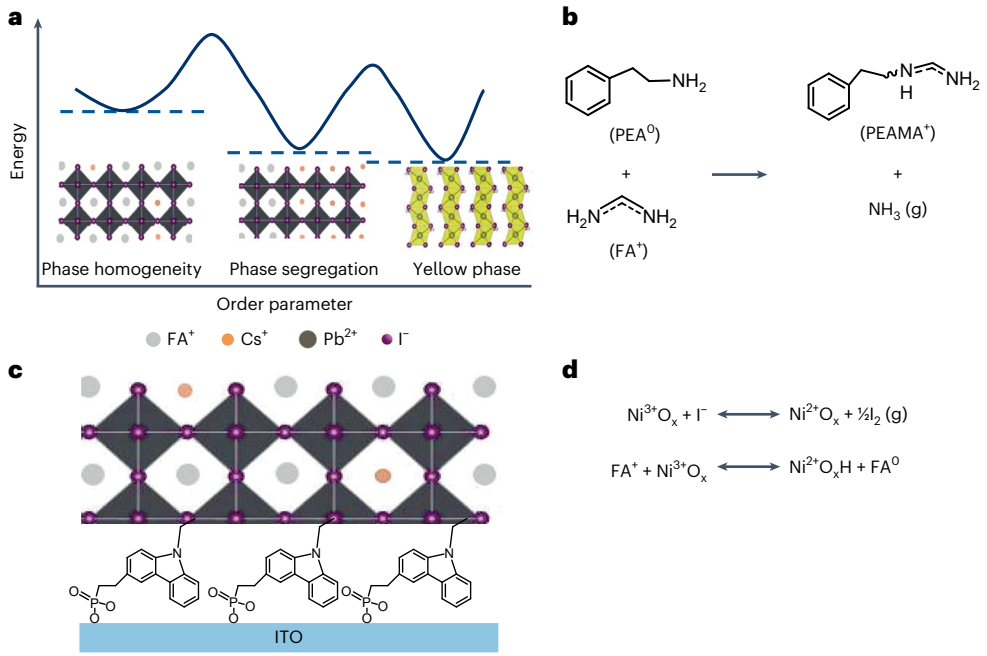

Fig. 3 | Design strategies to stabilize perovskite modules under thermal and chemical stress. a, Schematic of the decomposition pathways of perovskite layers under thermal stress. **b**, Thermally induced degradation of FA^+ at elevated temperature, with formation of volatile NH_3 gas and formamidine intermediates. **c**, Schematic of the possible interfacial contact between the perovskite layer, EtCz3EPA, and the ITO substrate, where the $-PO(OH)_2$ groups can bond with hydroxyl groups on the ITO surface, while the carbazole moieties interact with Pb^{2+} ions in the perovskite. **d**, Degradation of FA^+ and I^- at the interface between the NiO_2 layer and $FAPbI_3$, triggered by Ni_3^+ defect sites. In **a** and **c**, atom colour coding is used as follows: Pb, black; I, purple; FA^+, grey; Cs^+ orange.

75 °C, consistent with Goldschmidt tolerance factor predictions[61,62]. More research will be needed to determine how the percentage of Cs ions impacts material stability, with the goal of reducing Cs percentage, given that Cs is a rare-Earth element.

One might propose that the ideal composition for single-junction perovskite solar modules is pure $FAPbI_3$, as it is the simplest perovskite composition avoiding anion or cation segregation. However, the black phase of $FAPbI_3$, with its favourable bandgap (~1.48 eV) for record-efficiency devices, is not thermodynamically stable at room temperature or operating temperatures, and the presence of moisture makes it much more difficult to stay in the black phase[63,64]. The black phase of $FAPbI_3$ is generally obtained by strain engineering, either through templated growth on another material with a different lattice constant or by leaving volatile species such as methylammonium cation (MA^+) or additives that impose strain on $FAPbI_3$. Consequently, >1,000-h performance above T_{80} (the time a device maintains 80% of its initial performance under specific test conditions) of devices based on pure $FAPbI_3$ at 85 °C and constant illumination has been reported, indicating the promise of pure $FAPbI_3$ (refs. 35,65). One unanswered question is how long such strain can be maintained, particularly under high temperature and light and cycling thereof, when ion migration is considerably accelerated to release the strain. The tensile strain can be easily relaxed, whereas compressive strain might last longer. We note that higher temperatures (close to 100 °C) are more favourable to the black phase. Thus, one question is whether high temperature is a valid stressor for compositions with a stable black phase at high temperature but yellow phase at room temperature. Stabilizing the photoactive black phase of $FAPbI_3$ remains a central challenge in PSC research.

Chemical degradation of perovskites

Chemical reactions in perovskite devices are accelerated exponentially by high temperatures, following an Arrhenius process, and also by light-generated charges. Photochemical reactions involving iodide or other halide species are accelerated under high-temperature, full-spectrum sunlight exposure, as encountered under outdoor conditions. In addition, the A-site electrochemistry begins to appear. For example, organic cations can deprotonate and decompose at high temperatures. Compared to MA^+, FA^+ is less volatile, but it still undergoes thermal decomposition into ammonia (NH_3) and $(HCN)_x$ when temperatures exceed 95 °C. The Cs^+ cation, being much more stable under heat and light exposure, is often incorporated into FA-based perovskites to increase structural stability[36,66]. Additionally, multiple ammonium cations have been widely used to reduce trap density, discourage ion migration, and improve the structural stability of perovskites[67–69]. Unfortunately, during constant-illumination stress testing at high temperature, organic cations with low pK_a values, such as phenethylammonium (PEA^+, pK_a of 9.8), are prone to deprotonation to the amine. The resulting amine reacts with FA^+ to form new organic cations and NH_3, leading to perovskite degradation (Fig. 3b)[70]. Organic cations with high pK_a values, such as (phenethylamino) methaniminium cations ($PEAMA^+$, pK_a of 12), exhibit great potential for improving durability under constant illumination at 85 °C to ~1,500 h while operating above T_{80}. However, the long-term durability of these organic additives under extended testing remains uncertain. In addition, compositional tuning and crosslinking approaches enhance thermal stability, but may reduce charge extraction or introduce processing complexity, highlighting the balance between thermal resilience and optoelectronic performance. Mitigation strategies such as Cs-containing compositions and low-dimensional perovskite interface/additive engineering effectively slow photothermal degradation, but may reduce charge extraction efficiency, emphasizing the intrinsic trade-offs between stability and charge extraction[71,72].

Interface and charge-transport layer degradation

Higher temperatures and light also accelerate the degradation of CTMs and their interfaces with perovskites. Here we focus our discussion on CTMs that show record durability and are scalable. Polymer-based hole-transport materials (HTMs) such as

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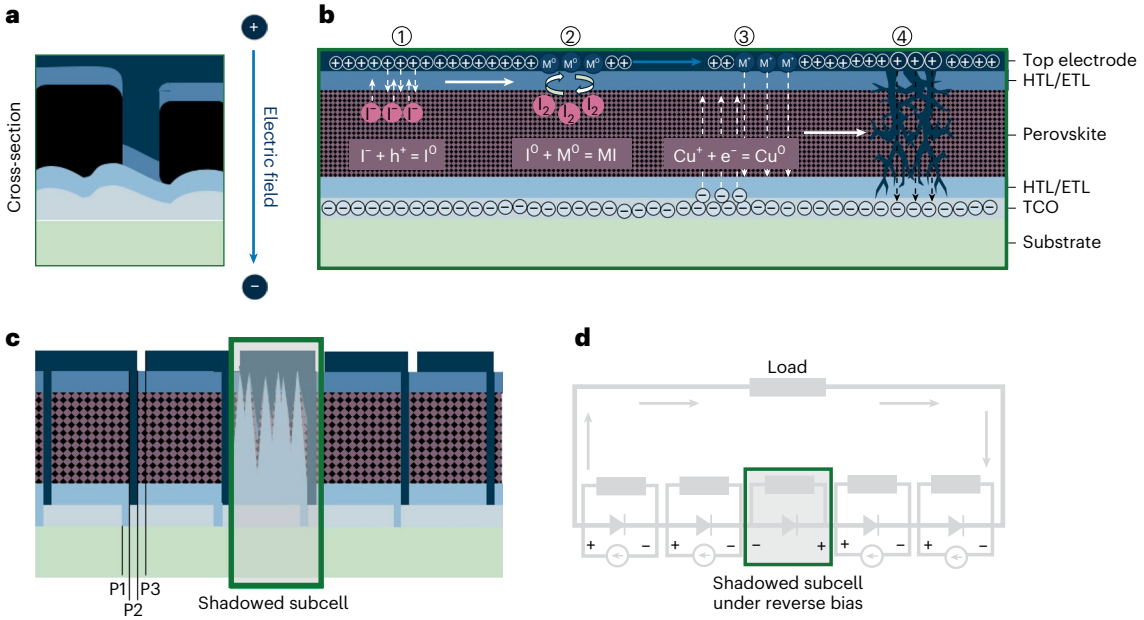

Fig. 4 | Illustration of degradation mechanisms under reverse bias and pinhole formation in PSCs. a, Cross-sectional representation of pinhole-formation-induced shunting. **b**, Sequential degradation reactions, including (1) iodide oxidation, (2) metal oxidation, (3) metal reduction and (4) conducting path formation. **c**, Partial-shading-induced reverse bias in subcells. **d**, Schematic of current flow under partial shading, where the shaded subcell acts as a load under reverse bias (note the inverted polarity relative to the illuminated cells). The uniform current through all subcells leads to local power dissipation and potential hotspot formation.

poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine] (PTAA) exhibit weak van der Waals interactions with perovskites, allowing quick A-site cation and halide migration at the interface. Although it has been reported to mainly reduce the UV durability of p–i–n devices, the weak interaction inevitably impacts thermal–light durability, because it is challenging to anchor the ions in perovskites onto TCOs[12]. Spiro-MeOTAD, another widely used HTM, faces severe challenges due to its low crystallization temperature. Recently, phosphonic-acid-based self-assembled molecules (SAMs) have emerged as a new category of HTM and have yielded highly efficient solar cells. However, the most popular SAMs, such as [2-(9H-carbazol-9-yl)ethyl]phosphonic acid (2PACz), are reported to be very unstable under high temperatures, which might be caused by the same weak interaction with perovskites, despite the fact that they stick well to indium tin oxide (ITO)[73]. New designs will be needed to make SAMs with functional groups to form strong bonds with both the substrate and the perovskite layer[12,74,75]. One notable example is [2-(9-ethyl-9*H*-carbazol-3-yl)ethyl]phosphonic acid (EtCz3EPA), as illustrated in Fig. 4c. The unique position and flexibility of the phosphonic acid linker group in EtCz3EPA enables face-on alignment of the carbazole group, dramatically increasing its interaction with perovskites.

Nickel oxide (NiO_x) is another promising HTM due to its excellent thermal durability. It can be deposited by both solution and physical vapour processes. In contrast to other oxides, such as MoO_3, its reaction with perovskites is much weaker, although not absent. Earlier studies showed that it easily deprotonates MA^+ cations, and a further study showed that it can also deprotonate FA^+ cations and oxidize iodine[76]. Figure 3d presents a schematic of such reactions. To mitigate these reactions, a physical barrier made of SAMs was used to improve the durability and compatibility of the NiO_x–perovskite interface[77,78]. Aluminium oxide or 1,3-bis(diphenylphosphino)propane deposited onto the NiO_x surface shields perovskites from deprotonation by NiO_x at high temperatures, with both achieving over 1,500-h T_{90} lifetimes at 85 °C under LED light and maximum power point tracking conditions[76,79]. However, these results have not been reproduced at the module level, indicating that the coverage of the SAM on NiO_x might not yet be well controlled.

The combination of bathocuproine (BCP) and fullerene is widely used in electron-transport layers in p–i–n for perovskite solar cells (PSCs). However, the mechanical weakness of fullerene imposes a limitation in terms of thermal cycling durability, and their non-compact packing also allows diffusion of I_2. In contrast, tin oxide (SnO_2) coated by ALD offers a substantially better barrier to I_2 and other species, such as metal ions from corroded electrodes. Replacing metal electrodes with TCO not only makes the devices semitransparent or bifacial, but also provides additional advantages such as avoiding corrosion of the metal electrode. A compact SnO_2 layer can protect the underlying layers from damage during TCO sputtering. ALD–SnO_2 is generally paired with fullerene, because directly depositing it on perovskites can damage the perovskites. Most of the reported ALD–SnO_2 depositions use tetrakis(dimethylamido)-Sn(IV) (TDMA-Sn) and moisture sources, carried out under low vacuum and elevated temperatures. The fullerene layer alleviates moisture damage during ALD deposition. However, the fullerene layer might not be compact enough to block moisture or TDMA-Sn. Given the very low pK_a of TDMA-Sn (~3.6), it can damage the FA-containing perovskites, as recently reported by in ref. 80. Further research is essential to redesign the ALD process or device stacking to minimize damage to the perovskite layer. Additionally, the conventional temporal ALD process for depositing a 20-nm SnO_2 layer typically requires more than 1 h, highlighting the need to shorten the deposition time. Further optimization may be achieved by employing alternative oxidants other than water, such as ozone, to increase the barrier properties of SnO_2 (ref. 81).

Reverse bias stability

Reverse bias, which can be caused by partial shading of solar modules during operation, partial module failure or a mismatched photovoltage output among modules, can pose a considerable threat to the longevity of thin-film PSCs[82]. It is of more concern for thin-film polycrystalline PSCs (not on single crystals) due to the smaller perovskite film thickness than in other types of solar cell, especially if not thin-film technologies, notorious ion migration under electric field, and easy electrochemical reaction of perovskites with nearly every type of metal

electrode. Even perovskite devices with reduced defect density and high crystallinity can remain vulnerable to electrochemical degradation under sustained high-bias conditions due to residual ionic mobility, interfacial reactivity or strain accumulation[83–85]. This issue has been studied less than other stressors. Earlier studies showed that PSCs could be damaged under reverse bias close to open-circuit voltage after only a few minutes[86]. Recently, very promising reverse-bias durability has been reported based on an improved understanding of device breakdown mechanisms.

An optimal thickness in the range of 1 µm or smaller makes perovskite films susceptible to pinhole formation during the film-coating process. Film pinholes can be caused by dust in the air, hydrophobic charge-transport layers (on which the perovskites are coated) that are too hydrophobic, inhomogeneous charge-transport layers, or non-uniform solvent drying and crystallization during perovskite coating[87]. Vacuum deposition might reduce the number of pinholes compared to solution processing, but there is insufficient data for this evaluation[88]. Carefully controlling the hole-transport layer and perovskite ink can greatly reduce pinhole formation. The presence of a relatively thick and less-conducting hole-transport layer can also mediate breakdown behaviour by blocking direct contact of the metal electrodes with TCOs, even in the presence of pinholes (Fig. 4a), as observed by Jiang and colleagues[89]. Electron-transport layers deposited by thermal evaporation should form a more continuous layer on perovskites. ALD is excellent in this context for blocking the direct contact of electrodes and thus avoiding breakdown.

Eliminating defects such as pinholes may reduce the chance of breakdown, but it cannot prevent PSC degradation. PSCs still degrade much more quickly under reverse bias than as a result of any other stressor. In one extreme example, very thick single-crystal perovskite devices without pinholes (made for ionization radiation) still feature a fast electrochemical reaction between perovskites and electrodes under a high-voltage bias[90]. Even without metal electrodes, the redox reaction of perovskite will still occur, as will ion migration under electrical biasing, as shown in Fig. 4b. Mitigating these issues can improve device reverse-bias durability, for example, by using a less-reactive metal than Cu[91], using a neutral carbon electrode[92] or increasing the perovskite crystallinity[93]. Among the various issues, the redox reaction related to iodide is the most challenging to overcome. Iodide can be reduced to neutral iodine or even positive iodide by injecting electrons under reverse bias, but this etches the metal electrodes and damage perovskites. Reducing the dark current by using stronger electron-injection buffer layers[94], reducing hole–electron injection so that the oxidation of iodide is not balanced[89], or even trapping the generated iodine[95] can improve reverse-bias durability.

Shadowing-induced degradation can be dramatically alleviated in multiple-junction monolithic tandem solar cells such as perovskite–silicon tandem solar cells due to the much more robust reverse-bias durability of bottom cells, as demonstrated by Di Girolamo and colleagues[96]. However, it can also be an issue for single-junction perovskite solar panels, including four-terminal tandem solar cells. Partial shading induces reverse-bias conditions within subcells, as shown in Fig. 5c,d. It should be noted that shadowing and reverse-bias stress tests follow different procedures because they serve different purposes. Most reported studies aim to limit degradation associated with partial shading and thus test devices under a fixed-voltage bias[16]. Despite the great progress made in improving the durability of single-junction solar cells to retain >80% of their performance for 1,000 h under reverse-bias conditions or open-circuit voltages, it is still an open question whether these measurements can evaluate the performance impact on partially shaded cells, where the accumulated voltages can be above hundreds of volts for modules connected in series. Ultimately, a combination of a bypass diode and improvements in the durability of cells under partial shading conditions is probably needed to solve this problem. Performance loss under reverse voltage has improved greatly in recent years, with protective layers preventing electrode damage, even under −15-V stress. However, this alone is insufficient. During partial shading, power dissipation in shaded cells can lead to high local temperatures depending on the module design. Therefore, materials and interfaces must be engineered to withstand both a high reverse voltage and elevated thermal stress. Similar hotspot mechanisms are expected in tandem architectures. Thermal simulations for tandem devices with efficiencies around 24% indicate that an optimized module design can limit hotspot temperatures to below ~86 °C under worst-case shading[97]. Reverse-bias degradation and potential-induced degradation (PID) are often confused, but they arise from distinct physical origins. PID originates at the module or system level, where several hundred volts (up to ±1,500 V) across the metal frame and solar cells cause alkali ions in glass to migrate towards the device interface, causing device degradation. Some recent studies have shown that high potential induces the rapid degradation of PSCs[98], with the accumulation of sodium ions at the glass/perovskite interface triggering the formation of an electron inversion layer at the interface[99]. In striking contrast, the presence of sodium ions on ITO has been reported to improve device efficiency due to passivation. Further studies are needed to reconcile this discrepancy. Nevertheless, using sodium-free glass or polyethylene terephthalate can avoid PID[100].

Degradation arising from module serialization

Upscaling from small-area PSCs to modules introduces unique durability challenges that are primarily linked to defective locations[101]. Figure 5a–c shows the structural differences between individual cells and modules, highlighting the additional complexities and processes needed for module fabrication. Defects such as voids in perovskite films and shunt sites under partial shading conditions act as initiation points for degradation that subsequently spreads across the module[102]. The extent and control of degradation spreading remain unclear and require further investigation.

The fabrication of monolithic modules typically relies on three laser scribing steps, P1, P2 and P3, to define the series interconnection[103]; in some advanced designs, an additional P4 step may be introduced for extended isolation or integration purposes. Misalignment or imprecise scribing damages the underlying layers, promoting defects and electrical shunts[104]. Specifically, P2 scribing can damage the perovskite films and can, if not properly managed, expose the perovskite directly to the metal electrodes, initiating degradation. P3-related delamination also increases exposure to environmental stressors, accelerating moisture ingress and mechanical failures[105]. These degradation mechanisms at laser scribing points are shown in Fig. 5f. Module encapsulation greatly reduces moisture ingress. However, if the encapsulation is imperfect or compromised, accelerated moisture ingress can occur, particularly around laser-scribed regions, as a thickness mismatch, cracks or delamination introduced by the laser processing can create localized weak points and pathways for moisture penetration. The entire dead region between P1 and P3 may suffer quick degradation because photogenerated charges are not extracted[106]. Preferential degradation occurs at the module interconnection sites, as illustrated in Fig. 5e. Wider interconnection zones can lead to increased inactive areas, negatively impacting device performance. Incorporating diffusion barrier layers (DBLs) is an effective strategy to prevent metal electrode diffusion and thereby enhance module durability[107–109].

One of the most-used materials for high-efficiency and long-term durability in small cells is C60[123]; however, when scaling up to modules, delamination and peeling problems occur over time. This peel-off issue is caused by thermal-expansion stresses at interconnections and scribe lines. In inverted architectures, adhesion problems, particularly at the top electrode interface, are often exacerbated by the poor adhesion of C_{60} to adjacent transport or electrode layers, making the stack

https://doi.org/10.1038/s41560-026-01969-z

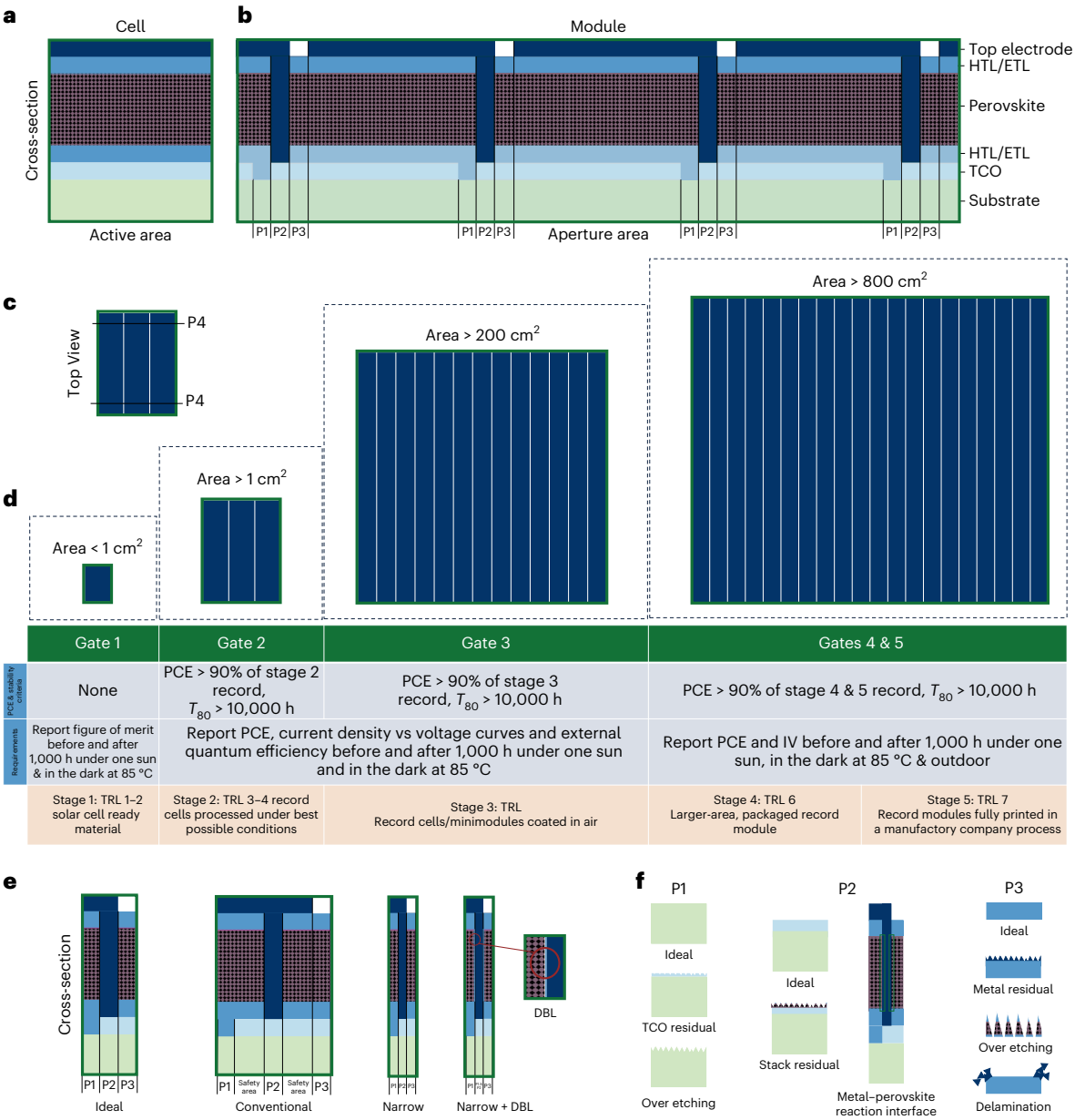

Fig. 5 | Challenges and potential solutions for perovskite module durability. **a**–**c**, Schematic representations of the cross-section of a cell (**a**) and a top view of a module (**b**), showing the additional processes needed for modules (P1, P2, P3). An optional P4 scribe, shown in **c**, may be introduced in advanced designs for edge isolation or rear contact segmentation. **d**, Indicative gate-stage development pathway linking laboratory research to pre-industrial validation. The sequence follows the logic used in EU and IEA PVPS programmes, where each gate corresponds to a defined Technology Readiness Level (TRL) milestone, from small-area devices (TRL 3–4) to pilot-scale prototypes (TRL 6–7) and field-tested demonstrators (TRL 8–9). **e**, Schematics of the ideal, conventional, narrow, and narrow with diffusion barrier layers (DBLs) interconnection part of perovskite modules. **f**, Ideal removal and principal degradation pathways at the interconnection of modules, occurring during the P1–P2–P3 scribing processes.

vulnerable to delamination and mechanical failure. Such adhesive failure of the C_{60} layer represents a form of physical degradation, and when it occurs near interconnects in modules, it can disrupt charge transport and lead to cascading failures. Even in encapsulated modules, continuous exposure to humidity and UV radiation can progressively degrade sealing materials and adhesives at interconnections and scribe lines, eventually compromising the encapsulation integrity and facilitating environmental ingress that accelerates overall module degradation. To improve module lifetime, future research could explore advanced materials with higher thermal and mechanical tolerance, optimized laser parameters to minimize damage during scribing, and innovative encapsulation techniques combining hybrid inorganic–organic layers for better adhesion and moisture protection. PIB edge sealing offers initial moisture protection but suffers from limited long-term reliability due to UV-induced degradation, deformation under thermal cycling, and insufficient mechanical robustness[110–112]. These weaknesses lead to progressive delamination and encapsulation failure, requiring advanced encapsulation strategies with superior thermal, mechanical and UV resistance. Encapsulation strategies that enhance environmental stability often compromise the overall cost of the device, optical transmission and flexibility, underscoring the need to optimize trade-offs between protection and device integration. The effective management of defects, degradation propagation, and encapsulation robustness is essential, but understudied. Adopting structured stage-gate processes may help systematically address these challenges (Fig. 5d)[113].

Outlook

Despite the diverse degradation pathways outlined in this Review, the outlook for achieving durable perovskite solar modules remains promising. Progress is advancing through targeted interface and composition engineering to suppress ion migration, strain management to prevent cracking, and robust encapsulation against environmental stressors. High-throughput degradation mapping and combinatorial approaches are accelerating material discovery and validation.

A structured framework is emerging. In the short term, reproducible fabrication and stability protocols are critical. In the midterm, validated encapsulation and upscaling strategies must be integrated. In the long term, predictive modelling linked to outdoor reliability data will enable realistic lifetime forecasting.

Future efforts should prioritize verified durability over efficiency gains, with improved device reproducibility, adopting standardized multi-stressor testing protocols. Although single-junction perovskites remain key for understanding degradation mechanisms, tandem architectures hold the strongest industrial potential. Ultimately, advancing both systems towards mechanically and chemically resilient devices capable of multi-decade field operation will be essential for the technology's commercial success.

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Acknowledgements

This work is supported in part by the Office of Naval Research (award no. N6833522C0122) and the UNC Research Opportunity Initiative. L.A.C. acknowledges funding from the European Union's Framework Programme for Research and Innovation Horizon Europe (2021–2027) under Marie Skłodowska-Curie grant agreement no. 101068387 'EFESO'. This work was authored in part by the National Laboratory of the Rockies for the US Department of Energy (DOE), operated under contract no. DE-AC36-08GO28308. Funding was provided in part by the 'Perovskite Enabled Tandems' project funded by the US DOE Office of Energy Efficiency and Renewable Energy EERE and Solar Energy Technologies Office (award #52776). The work at CubicPV was supported by the US DOE under awards nos. DE-EE0011421 and DE-EE0009529. The views expressed in this Review do not necessarily represent the views of the DOE or the US Government. The US Government retains and the publisher, by accepting the Article for publication, acknowledges that the US Government retains a nonexclusive, paid-up, irrevocable, worldwide license to publish or reproduce the published form of this work, or allow others to do so, for US Government purposes.

Author contributions

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Competing interests

J.H. is the founder of Perotech. D.B.D. is the director of research and development at CubicPV Inc. The remaining authors declare no competing interests.

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Peer review information *Nature Energy* thanks Marc Koentges and Jianfeng Lu and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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